

KU Leuven
Laboratory for Neuro- and Psychophysiology
Campus Gasthuisberg, O&N 2
Herestraat 49 bus 1021
BE-3000 Leuven
Belgium
arne.vandenkerchove@kuleuven.be
+32 473 32 78 71

June 11, 2025

Journal of NeuroEngineering and Rehabilitation
Springer Nature group

Dear editors of the Journal of NeuroEngineering and Rehabilitation,

We are pleased to submit our manuscript entitled “Visual ERP-based brain-computer interface use with severe physical, speech and eye movement impairments: case studies” for consideration to publish in your journal.

This study is situated at the intersection of neuroscience, biomedical engineering, and physical medicine and rehabilitation, and is therefore well aligned with the scope of JNER. It directly contributes to ongoing efforts to advance assistive technologies that support communication in individuals with severe motor and speech impairments, especially those with limited or no eye motor control. The work proposes, implements, and evaluates a brain-computer interface (BCI) with immediate clinical relevance for communication rehabilitation.

The manuscript presents a series of case studies involving individuals with severe speech, physical, and gaze impairments (SSPGI), evaluating the feasibility and usability of a gaze-independent visual oddball BCI. We assessed performance across three visuospatial attention (VSA) conditions – overt, covert, and free VSA – in participants with varying levels of gaze control. Our results confirm that while covert attention with cued central fixation leads to reduced accuracy, free VSA, where gaze is unconstrained, often performs on par with overt VSA. Importantly, we demonstrate that decoder training under overt VSA conditions can improve subsequent decoding performance in covert or free settings. These findings suggest that residual eye motor control can be leveraged to improve BCI operation, even in individuals with severe impairments.

This work addresses a major gap in the literature by focusing on realistic users with SSPGI, rather than healthy controls. It offers actionable insights into the adaptation of visual BCI protocols for user comfort and performance, highlights the importance of user-centered design, and proposes recommendations for future clinical implementation. The study also provides a combination of detailed clinical, neurophysiological, and behavioral data in this understudied

population.

We believe our findings will be of interest to your readership, particularly those working on novel BCI paradigms, clinical neurorehabilitation technologies, and translational research in assistive communication.

Thank you for considering our manuscript for publication in Journal of Neuro-Engineering and Rehabilitation. We look forward to your response.

Sincerely and on behalf of all authors,

A handwritten signature in black ink, consisting of several overlapping loops and a long, sweeping stroke extending towards the upper right.

Dr. ir. Arne Van Den Kerchove